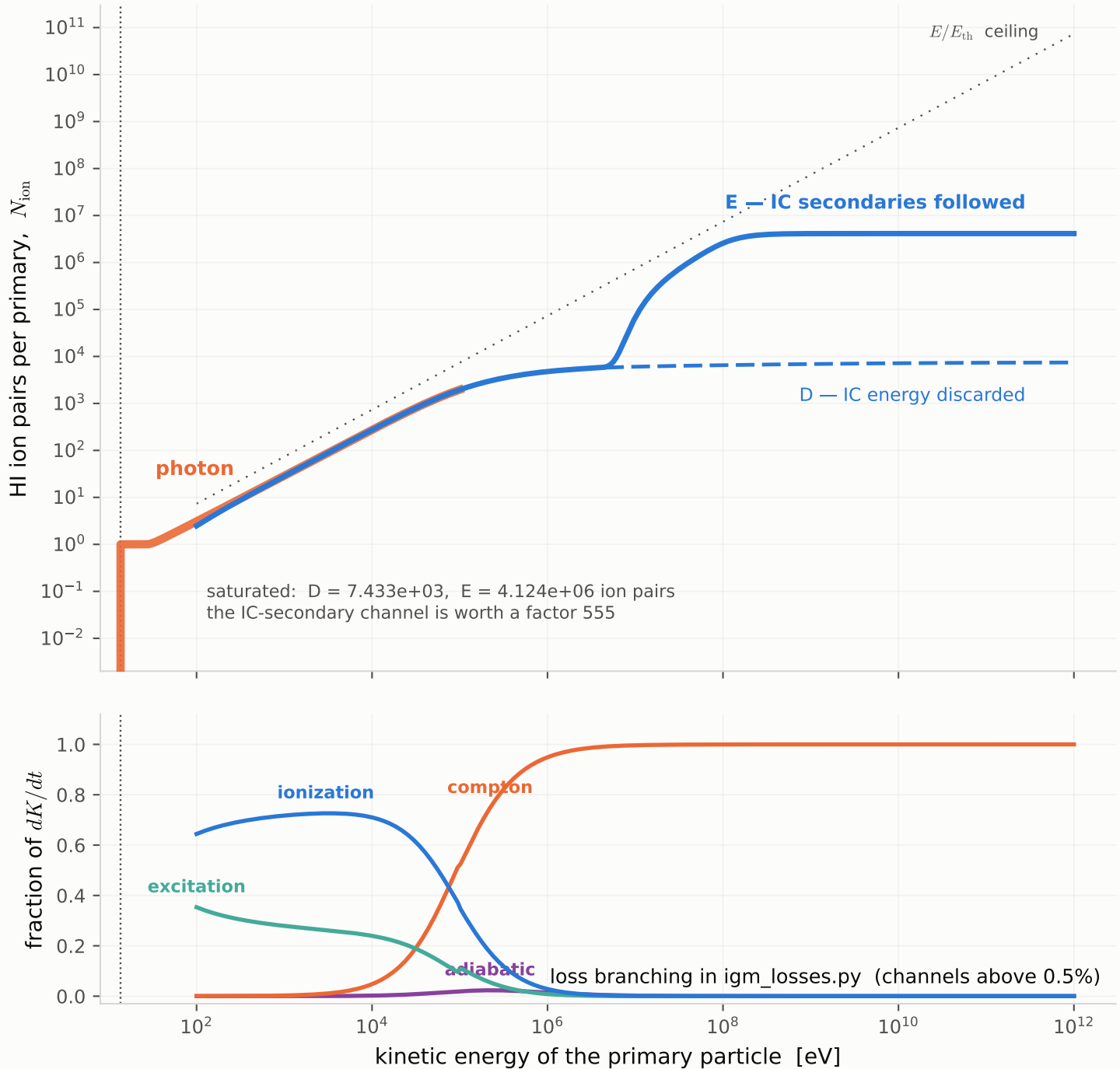


Average HI ionizations per primary particle — igm_losses prescription IGM at $z = 20$, $x_e = 10^{-4}$, all cascade generations counted



Electron propagation: igm_losses.py — adiabatic, synchrotron, IC with Klein-Nishina, Coulomb (Gould 72), excitation (Stone & Kim 02), ionization (RBED, Kim+00), bremsstrahlung. Ionizations counted event by event; knock-on electrons followed through the BED secondary spectrum (Kim & Rudd 94). Photon yield on the same prescription. $n_{\text{HI}} = 1.7595\text{e-}03 \text{ cm}^{-3}$, $B_0 = 1.0\text{e-}13 \text{ T}$, $H(z) = 1.1802\text{e-}16 \text{ s}^{-1}$ — all adjustable via igm_config.